

Arabic Text-to-Speech (TTS) System Using YouTube Data and Deep Learning

Project Description:

This project focuses on building a high-quality Arabic Text-to-Speech (TTS) system by scraping YouTube videos, transcribing the audio (if necessary), and training a TTS model using **Tacotron 2** (or a similar model). The goal is to create a natural-sounding Arabic speech synthesis system, with a focus on Egyptian Arabic but allowing for multiple dialects. Data will be gathered from male and female voices on diverse topics to ensure high variability and better model generalization. The final deliverable includes a trained model, evaluation results, and an IEEE-style paper with a project presentation.

Requirements:

1. Scrape a minimum of **500 Arabic YouTube videos**, each with a duration of **5 to 15 minutes**.
2. **Transcribe the audio** using OpenAI Whisper, but only if no transcripts are provided with the videos.
3. **Data preprocessing:** Clean, label, and organize the dataset for TTS model training.
4. Extract **audio embeddings** using the **Wave2Vec** model (recommended).
5. Train a **Tacotron 2-based** TTS model (alternative models can be considered).
6. Evaluate the model on various criteria such as **speech clarity, naturalness, and accuracy on untrained topics**.
7. **Bonus:** Deploy the final system using **Flask** or **FastAPI** for online access.

Data Gathering Protocol:

- **Source:** Scrape Arabic speech data from YouTube across a variety of topics (e.g., news, education, entertainment).
- **Diversity:** Collect audio data with both **male and female voices** across multiple **Arabic dialects**, with a focus on **Egyptian Arabic**.
- **Duration:** Audio files should be between **5 to 15 minutes** long.
- **Transcription:** Use **OpenAI Whisper** to transcribe audio if necessary, or rely on provided transcripts.

Timeline:

- **Phase 1: Data Cleaning & Preparation (12th November)**
 - Scrape YouTube videos, transcribe audio, and preprocess the data for model training.
- **Phase 2: Model AB Testing (2nd December)**

- Train and evaluate multiple TTS model configurations, focusing on speech quality and generalization.
- **Phase 3: IEEE Paper & Presentation (22nd December)**
 - Present a 10-slide summary and submit an IEEE-style paper, followed by team discussion and feedback.

Deliverables:

Phase 1: Data Collection & Preparation

- **Scraping Script:**
 - A Python script using **YouTube Data API** or **youtube-dl** that scrapes at least 500 Arabic YouTube videos.
 - Error handling and metadata extraction (title, duration, etc.).
- **Audio Transcription:**
 - Use **OpenAI Whisper** to transcribe audio when necessary.
 - Deliver a dataset with paired audio and transcriptions.
- **Preprocessed Dataset:**
 - Clean, labeled, and formatted dataset ready for training.

Phase 2: Model Training & Testing

- **TTS Embeddings & Trained Model:**
 - Extract audio embeddings using **Wave2Vec** (recommended).
 - Train a **Tacotron 2-based** model using the processed dataset.
 - Deliver **trained model weights** (in .h5 format) and the training code.
- **Testing Report:**
 - Evaluate the model's performance for speech clarity, naturalness, and accuracy, especially on untrained topics.
 - Perform AB Testing to compare different model configurations.

Phase 3: Final Documentation & Presentation

- **IEEE-Style Final Project Report:**
 - A report detailing the project objectives, data gathering process, model training, and evaluation.
 - Include scripts, models, and instructions for replicating the project on **GitHub**.
- **10-Slide Presentation:**
 - Summarize the project outcomes in a 10-slide presentation to be discussed on **22nd December**.
- **Bonus:** Deploy the system using **Flask** or **FastAPI** (if applicable).